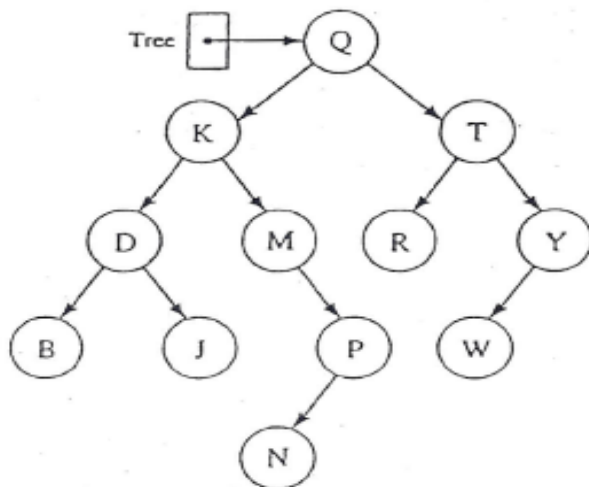


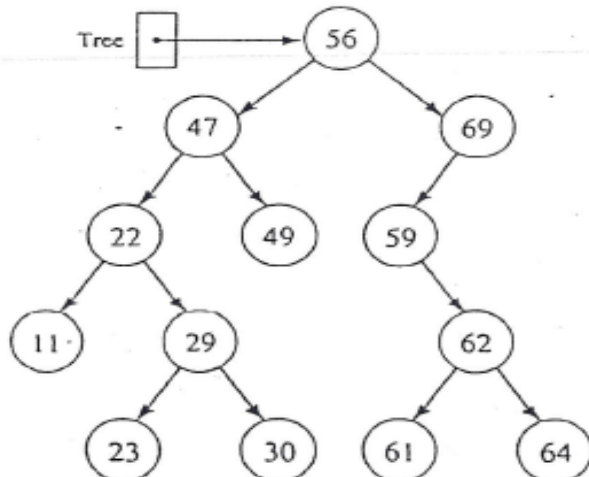
CSC310 Data Structures and Algorithm Analysis (Dr. Ng) Homework 2

Note the question numbers are from 8-19 and abide to the standard of Homework Assignment.



8. Show what the tree would look like after each of the following changes. (Use the original tree to answer each part.)
 - (a) Add node C.
 - (b) Add node Z.
 - (c) Add node X.
 - (d) Delete node M.
 - (e) Delete node Q.
 - (f) Delete node R.
9. Show what would be printed by each of the following.
 - (a) an inorder traversal of the tree
 - (b) a postorder traversal of the tree
 - (c) a preorder traversal of the tree
10. Draw the binary search tree whose elements are inserted in the following order:
50 72 96 94 107 26 12 11 9 2 10 25 51 16 17 95

Exercises 11–15 use the following tree.



11. (a) What is the height of the tree?
 (b) What nodes are on level 3?
 (c) Which levels have the maximum number of nodes that they could contain?
 (d) What is the maximum height of a binary search tree containing these nodes? Draw such a tree.
 (e) What is the minimum height of a binary search tree containing these nodes? Draw such a tree.
12. (a) Trace the path that would be followed in searching for a node containing 61.
 (b) Trace the path that would be followed in searching for a node containing 28.
13. Show what would be printed by each of the following.
 (a) an inorder traversal of the tree
 (b) a postorder traversal of the tree
 (c) a preorder traversal of the tree
14. Show how the tree would look after the deletion of 29, 59, and 47.
15. Show how the (original) tree would look after the insertion of nodes containing 63, 77, 76, 48, 9, and 10 (in that order).
16. True or False.
 (a) Invoking the delete procedure in this chapter might create a tree with more levels than the original tree had.
 (b) A preorder print traversal will print the exact reverse of what a postorder traversal will print.
 (c) An inorder traversal will always print the elements of a tree in the same order, regardless of the order in which the elements were inserted.
 (d) A preorder traversal will always print the elements of a tree in the same order, regardless of the order in which the elements were inserted.
17. If you wanted to traverse a tree, writing all the elements to a file, and later (the next time you ran the program) rebuild the trees by reading and inserting, would an inorder traversal be appropriate? Why or why not?
18. (a) One hundred integer elements are chosen at random and inserted into an ordered linked list and a binary search tree. Describe the efficiency of searching for an element in each structure, in terms of Big-O.
 (b) One hundred integer elements are inserted in order, from smallest to largest, into an ordered linked list and a binary search tree. Describe the efficiency of searching for an element in each structure, in terms of Big-O.
19. The Info field of the nodes in a binary search tree contains short character strings.
 (a) Show how such a tree would look after the following words were inserted (in the order indicated):
 monkey canary donkey deer zebra yak walrus vulture penguin quail
 (b) Show how the tree would look if the same words were inserted in this order:
 quail walrus donkey deer monkey vulture yak penguin zebra canary
 (c) Show how the tree would look if the same words were inserted in this order:
 zebra yak walrus vulture quail penguin monkey donkey deer canary